Fault-Tolerant Quantum Portfolio Optimizer with Dynamic Constraints

Logical-qubit resource estimation, error-mitigation ablation, significance tests and a live dashboard

SYNTHETIC-DATA DRAFT. Every number that depends on market data (sections 7-9: ablation instances, backtests, scaling instances) comes from a simulated market, because the build environment had no internet access (Yahoo/FRED/ESG could not be downloaded). They illustrate the pipeline and must not be read as results on real markets. Sections 5-6 (surface-code simulation, resource estimates) and the theory in section 9 do not depend on market data. To produce the final report run `python report/build_report.py --results results_real` after the real-data run: this banner disappears automatically only when `meta.json` records the real-data source.

Abstract. We build a hybrid quantum-classical, walk-forward portfolio optimizer whose multi-period decision problem is a time-dependent QUBO (time-varying risk aversion and weight caps driven by VIX, transaction-cost surrogate, hard ESG exclusion and soft ESG tilt). The QUBO is solved by a layer-wise adaptive QAOA on an exact statevector simulator, under (i) a phenomenological NISQ+ noise model with dynamical decoupling (DD), zero-noise extrapolation (ZNE) and probabilistic error cancellation (PEC) and (ii) a logical-qubit mode whose logical error rates are extracted from Stim/PyMatching surface-code simulations. We compare with exact search, simulated annealing, a discrete MIQP, CVaR stochastic programs, and RL agents on out-of-sample windows containing the 2008 and 2020 crises. **The claim we defend is deliberately modest: rigorous bounds plus an empirical test of advantage criteria, not a proven quantum advantage.**

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1 Introduction and what is (and is not) claimed

The task statement asks to "prove (or rigorously bound) a quantum advantage in solution quality or sample complexity under realistic NISQ+ noise". No proof of an advantage of constant-depth QAOA over the best classical solvers on portfolio QUBOs is known, and our instances ($n \leq 16$ qubits) are solved exactly by enumeration in under a millisecond, so any advantage could only be a *scaling* statement. We therefore contribute: (1) exact bounds for our noise model (success-probability lower bound, PEC/ZNE sampling costs, a noise budget that any sample-complexity advantage must respect); (2) an empirical scaling comparison for $n \leq 16$; (3) logical-qubit resource estimates with an explicit assumption table and sensitivity analysis; (4) an out-of-sample statistical comparison with multiple-testing correction; (5) a fully reproducible pipeline. Where the data are negative we report them as negative.

Contributions

- Multi-period time-dependent QUBO with hard and soft constraints, verified against the direct objective (unit tests) and against a hard-budget MIQP formulation.
- QAOA with DD, ZNE, PEC and a logical-qubit mode; ablation of each mitigation (section 7).
- Resource estimator with consistent time/failure model, magic-state error check, and sensitivity sweep over every constant (section 6).
- Walk-forward backtests with stationary-bootstrap Sharpe tests, Wilcoxon tests and Holm correction (section 8).

2 Related work

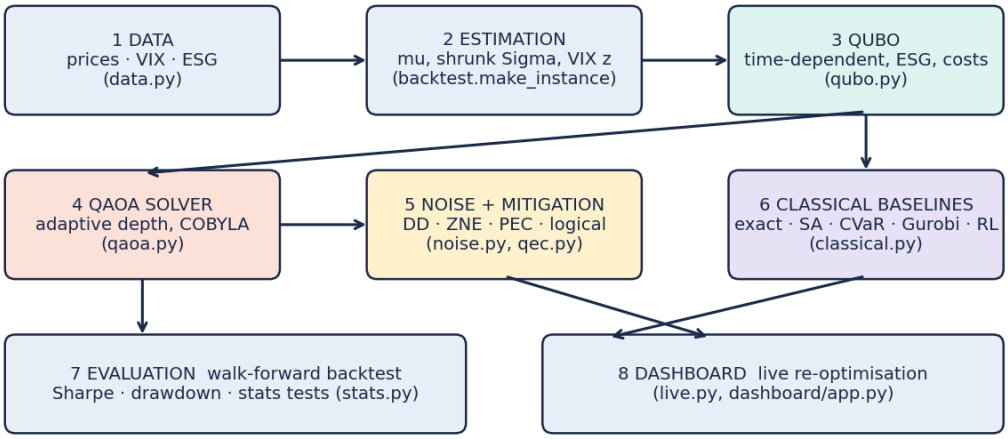
Walk-forward QUBO rebalancing with QAOA on classical simulators (Weinberg, arXiv:2603.16904); multi-period annealing with quadratic transaction-cost surrogates (arXiv:2607.03218); multi-period QUBO with market impact (Rosenberg et al., IEEE JSTSP 2016); the closest study of advantage under noise for portfolio QAOA (Yuan et al., arXiv:2410.16265), which finds that realistic noise precludes advantage except when measurement noise dominates; QAOA itself (Farhi et al., arXiv:1411.4028); error mitigation (Temme-Bravyi-Gambetta, arXiv:1612.02058); noisy-device limitations (Stilck França & García-Patrón, Nat. Phys. 2021); surface codes and lattice surgery (Fowler et al. 2012; Fowler & Gidney, arXiv:1808.06709; Litinski, Quantum 2019); Stim (Gidney, Quantum 2021) and PyMatching v2 (Higgott & Gidney, arXiv:2303.15933); rotation synthesis (Ross & Selinger, arXiv:1403.2975; Bocharov et al., PRL 114, 080502). *The entries for Weinberg, arXiv:2607.03218 and Yuan et al. were located by search during project setup and the others cited from memory; verify every citation before submission.*

3 Problem formulation

At each rebalance date t (every 21 trading days) we use only the previous 252 days. Assets below the ESG threshold (40.0) are excluded (**hard** constraint). From the eligible set, 4 assets are selected greedily (trailing Sharpe, penalising correlation) to fit on qubits. Each asset-period weight is $w = \text{level}/6$ with $\text{level} = b_0 + 2\,b_1$ ($K = 2$ bits), over 2 look-ahead periods, so $n = A \cdot K \cdot T = 16$ qubits. The energy to minimise is

$$E(x) = \Sigma_t \left[-\mu_t \cdot w_t + (\lambda_t/2)\, w_t' \Sigma_t\, w_t + c_{tc} \,\|w_t - w_{t-1}\|^2 + P_b (1' w_t - 1)^2 + P_c \cdot \text{cap}_t \text{ penalty} - \tau\, s \cdot w_t/100 \right]$$

Time-varying constraints. With z the 252-day z-score of the previous day's VIX: $\lambda_0 = \lambda(1 + \kappa\, \text{clip}(z, -1, 3))$ with $\lambda = 6.0$, $\kappa = 0.5$; the weight cap drops from 3/6 to 2/6 when $z > 1.0$. **Transaction costs**: linear 10.0 bp in the P&L, quadratic surrogate $c_{tc} = 0.02$ inside the QUBO. **ESG**: soft tilt $\tau = 0.002$. The soft budget/cap penalties are followed by a hard repair (non-negative, fully invested). The QUBO matrix is checked against the direct objective for random bitstrings (tests/test_core.py), and its exact optimum equals the optimum of the hard-budget MIQP whenever the penalty is large (tests/test_miqp.py).



Pipeline architecture (data, estimation, QUBO, solvers, noise and mitigation, evaluation, dashboard).

3.1 Data, universe and audit trail

Synthetic market: 16 assets in 4 sectors with a market factor, sector factors and idiosyncratic Student-t noise (6 d.o.f.), a two-state volatility regime (x2.8 in stress) forced on to the 2008-09 and 2020 crisis windows, a VIX proxy driven by the regime, and random ESG scores in [20, 95]. It exists to test the pipeline; it contains no information about real markets.

Look-ahead controls. Decisions at index i use returns and VIX strictly before i; macro series are forward-filled only and a series with insufficient coverage is dropped, never back-filled (FRED now truncates ICE-BofA series to three years, so back-filling the credit spread would copy 2023 values into 2008). Hyper-parameters for each test window are chosen on earlier data only (scripts/tune.py).

4 Algorithms and baselines

QAOA. Exact statevector simulation of p alternating cost phases $\exp(-i\gamma\pi E_{norm})$ and RX mixers; parameters by COBYLA, starting at p = 1 and adding a layer (up to $p_{max} = 3$) only while the relative energy improvement exceeds 1% (*adaptive ansatz*). The best of 2048 samples, evaluated classically, is decoded and repaired. Parameters are trained noiselessly and deployed under each noise model (noise-aware training appears only in the ablation).

Classical baselines on the same problems and out-of-sample periods.

- *exact_qubo* (true QUBO optimum, ceiling for every QUBO solver) and *sa_qubo* (single-flip simulated annealing).
- *miqp_sub*: the hard-budget discrete multi-period problem solved exactly (Gurobi MIQP via the matrix API when gurobipy is present, otherwise an exact dynamic program; identical optimum, cross-tested). *miqp_gurobi_full*: same on all ESG-eligible assets.
- *cvar_full / cvar_sub / cvar_gurobi_sub*: single-period $CVaR_{0.9}$ scenario programs (HiGHS and Gurobi) with 50% cap and linear costs - the stochastic-programming baseline; *_sub* uses the same assets as the QUBO, *_full* all eligible assets, separating selection from solver effects.
- *rl_es_full*: small MLP policy trained by evolution strategies (NumPy); *rl_torch_full*: PyTorch REINFORCE with a per-state baseline (same reward). Both train only on data preceding the test start.
- *equal_weight_full*.

```
for each rebalance date i in test window:                  # every 21 (or 63) trading days
    window = returns[i-252 : i]; z = vix_zscore(i-1)         # past data only
    elig = {assets with ESG >= esg_min}                      # hard filter
    sub  = greedy_select(window[elig], A)                    # assets that go on qubits
    Q    = build_qubo(mu, Sigma, w_hold, lambda_t(z), caps_t(z), esg)
    for method in methods:                                     # each keeps its OWN holdings
        x = solve(method, Q | scenarios | policy)            # QAOA / exact / SA / MIQP / CVaR / RL
        w = repair(first_period_weights(x))                    # hard constraints
        pay linear cost on |w - w_hold|; hold for 21 days; drift weights
```

Methods in this run: equal_weight_full, cvar_full, cvar_sub, miqp_sub, rl_es_full, exact_qubo, sa_qubo, qaoa_ideal, qaoa_noisy_raw, qaoa_noisy_mitigated, qaoa_logical. Optional packages available at run time: none of torch/gurobipy; missing optional methods are **absent from all tables**, not simulated.

5 Noise model and fault tolerance

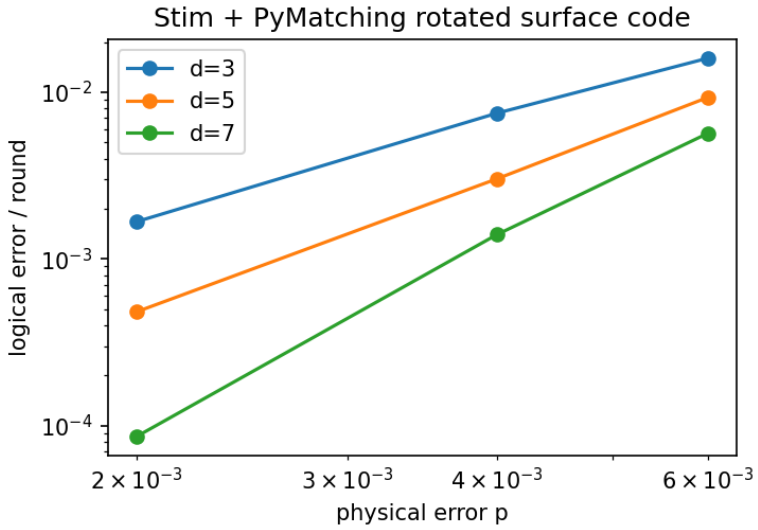
NISQ+ model. Gate counts (2 CNOT + 1 RZ per ZZ coupling per layer, RZ per linear term, RX mixer) are converted to an effective independent bit-flip probability q per qubit on the measured string: two-qubit error 0.003, one-qubit 0.0003, readout 0.01, idle 0.002 per qubit-layer. DD removes a fraction 0.9 of the idle error. This keeps the simulation exact and makes ZNE (noise scaling of the gate part) and PEC (inverse flip channel) analysable, but it is a **phenomenological** model: real devices have correlated and coherent noise.

Formally, with $N_{cx} = 2\, n_{zz}\, p$ CNOTs and $N_{rz} = (n_{zz} + n_{lin} + n)\, p$ rotations: $\lambda = (2/3)[p_2\, N_{cx} + p_1\, N_{rz} + r_{idle}(1–DD)\, n\, p]/n \cdot s$, $q_{gate} = (1 – e^{-2\lambda})/2$, $q = q_{gate} + r_{ro} – 2\, q_{gate}\, r_{ro}$, where s is the ZNE noise-scale factor (readout is not scaled) and the noisy distribution is $P_{noisy} = (\mathrm{flip}_q)^{\otimes n}\, P_{ideal}$. In logical mode q follows the same formula with $\lambda = –½\, \ln(1 – 2\, q_L)\, p\, s$ and no readout term.

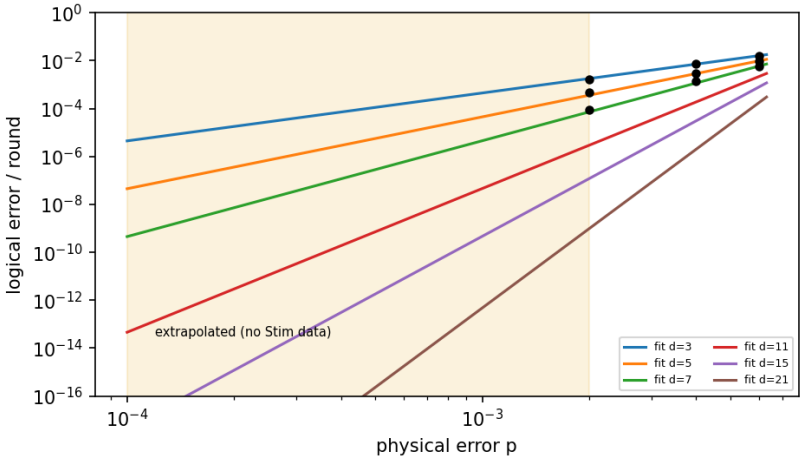
Logical qubits - what is and is not simulated. Stim simulates the rotated surface code *memory* experiment (d rounds, circuit-level depolarising noise, MWPM decoding by PyMatching). QAOA contains non-Clifford rotations and a 2^n statevector, which cannot be co-simulated at the code level. We therefore fit the Stim logical error rate and **inject it as an effective logical bit-flip probability into the small QAOA simulation**. That is a modelling assumption, not an end-to-end fault-tolerant simulation.

d	p	shots	fails	p_L_per_round	stderr
3	0.002	30000	75	0.00167	0.000288
3	0.004	30000	334	0.00748	0.000606
3	0.006	30000	708	0.016	0.000876
5	0.002	30000	36	0.00048	0.0002
5	0.004	30000	225	0.00302	0.000498
5	0.006	30000	684	0.00929	0.000862
7	0.002	30000	9	8.57e-05	0.0001
7	0.004	30000	146	0.0014	0.000402
7	0.006	30000	584	0.00566	0.000798

Stim + PyMatching memory sweep. Fit p_L/round = A $(p/p_{th})^{(d+1)/2}$: A = 0.044, p_{th} = 0.0099 (9 points; rows with <5 failures are excluded from the fit). Only p = 2e-3..6e-3 was simulated; p ≤ 1e-3 is an extrapolation.



Logical error per round vs physical error for d = 3, 5, 7 (Stim, memory only).



Fitted logical error per round vs p for larger distances, with the Stim data points (black). Below p = 2e-3 the curves are extrapolated.

With d = 7 and p = 0.001, the fitted model gives a logical flip probability of 1.6e-05 per gate layer for a Z-basis measurement. Note this is an extrapolation of a 9-point fit whose lowest-error points (d = 7, p = 2e-3) have only a handful of failures.

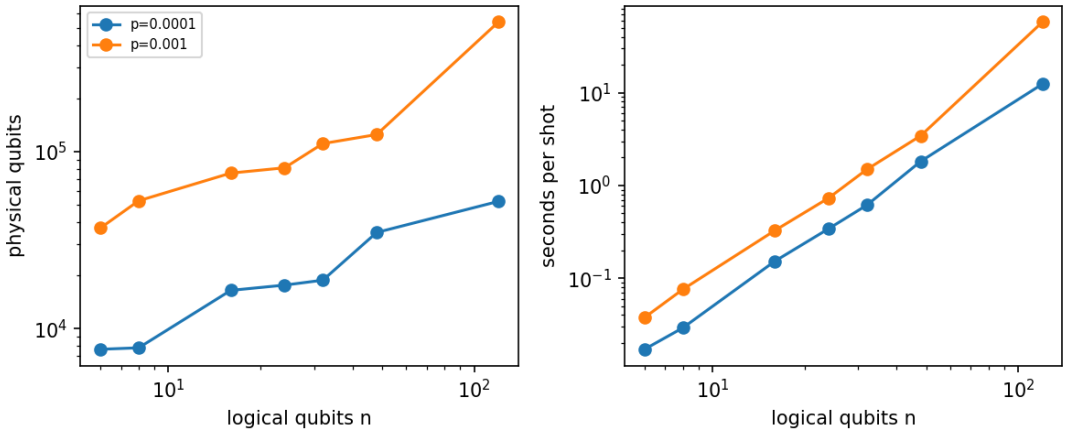
6 (a) Logical-qubit resource estimation

For each circuit we count rotations, synthesize them into T gates, and choose the smallest odd distance d such that the total failure budget (split in three equal parts: logical errors of the data patches, rotation-synthesis error, magic-state infidelity) is met. A single exposure time (the larger of the Clifford depth and the T-supply time) is used for both the failure estimate and the runtime. Magic states come from 15-to-1 distillation (1 or 2 levels, whichever meets the budget). Physical qubits = $n_{logical}$ · routing · $(2d^2–1)$ + factories · footprint · d^2.

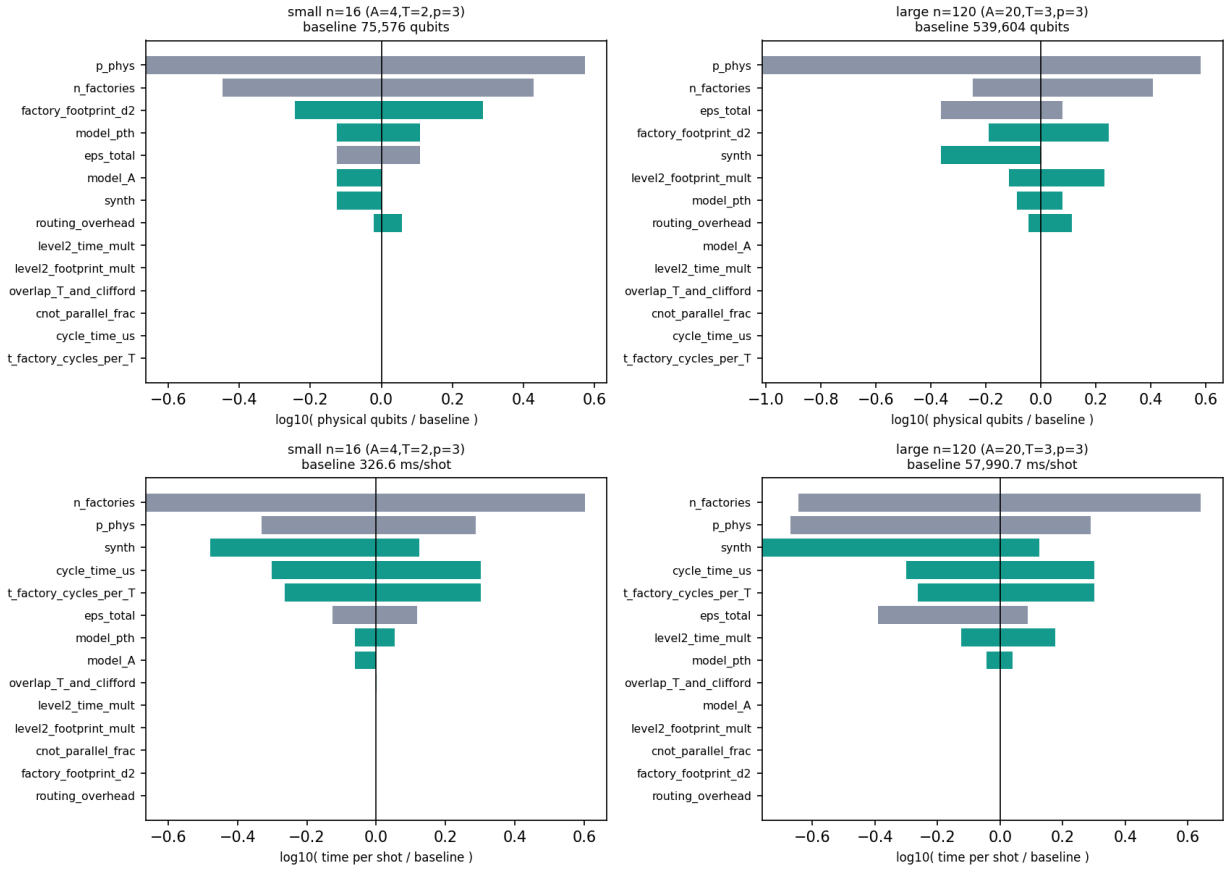
circuit	p_phys	scenario	distance	physical_qubits	time_per_shot_ms	distill_levels
small n=16	0.001	optimistic	9	49232	5.12	1
small n=16	0.001	baseline	15	75576	327	1
small n=16	0.001	pessimistic	19	88758	7.39e+03	1
small n=16	0.0001	optimistic	5	15184	2.85	1
small n=16	0.0001	baseline	7	16440	152	1
small n=16	0.0001	pessimistic	9	19878	3.5e+03	1
large n=120	0.001	optimistic	15	183480	541	1
large n=120	0.001	baseline	21	539604	5.8e+04	2
large n=120	0.001	pessimistic	27	1180620	1,939,098	2
large n=120	0.0001	optimistic	7	39864	253	1
large n=120	0.0001	baseline	9	52308	1.24e+04	1
large n=120	0.0001	pessimistic	13	146670	311,250	1

Optimistic / baseline / pessimistic corners (all constants moved together, plus fit-uncertainty multipliers on A and p_{th}). time = one circuit execution (one shot).

Reading. At p = 1e-3 the 16-qubit instance needs 75,576 physical qubits at baseline (range 49,232 to 88,758) and 327 ms per shot; a 120-qubit instance needs 539,604 (range 183,480 to 1,180,620) with 2-level distillation. Runtime is T-supply bound (the sweep shows cycle time, T cycles and synthesis cost dominate, routing hardly matters). A 16-qubit problem that a laptop solves in microseconds would thus need ~10^4-10^5 physical qubits for fault-tolerant execution: logical qubits make QAOA possible at scale, not competitive at this scale.



Baseline physical qubits and seconds per shot vs number of logical qubits (QAOA depth p = 3, dense couplings, constants at baseline).



Sensitivity of physical qubits (top) and time per shot (bottom) to each constant, one at a time (teal: modelling assumption, grey: design variable).

parameter	baseline	swept	source / justification	confidence
p_phys	0.001	0.003, 0.001, 0.0003, 0.0001	Design variable. Fit data only cover p=2e-3..6e-3, so p<=1e-3 is an EXTRAPOLATION.	low (extrapolated)
synth	RS 3*log2(1/e)	RUS 1.15*log2(1/e), RS 3*log2(1/e) [base], RS 3*	Ross-Selinger arXiv:1403.2975 (3 log2 + O(loglog), ancilla-free); RUS slope 1.15: Bocharov et al. PRL 114, 080502. The '+9.2' of the first draft was n	medium
routing_overhead	1.5	1.0, 1.5, 2.0, 3.0	Typical layouts use ~1.5-2x (Litinski, Game of Surface Codes); dense all-to-all ZZ couplings are routing-heavy.	medium
n_factories	4	1, 2, 4, 8, 16	Design choice; trades qubits for time.	n/a (design)
factory_footprint_d2	72.0	36.0, 72.0, 150.0	Order of 10^1-10^2 d^2 for 15-to-1 factories (Litinski, Magic state distillation: not as costly as you think). Not re-verified.	low
t_factory_cycles_per_T	5.5	3.0, 5.5, 11.0	Order-of-magnitude from the same references; layout dependent.	low
cycle_time_us	1.0	0.5, 1.0, 2.0	Superconducting-like ~1 us; trapped ions/atoms are 10-1000x slower.	medium (platform)
cnot_parallel_frac	1.0	0.25, 0.5, 1.0	Dense couplings serialise on a 2D layout; 1.0 is the optimistic end.	low
overlap_T_and_clifford	True	True, False	True is standard in game-of-surface-codes style schedules; False is the sequential upper bound.	medium
eps_total	0.05	0.01, 0.05, 0.2	Design variable.	n/a (design)
level2_footprint_mult	3.0	2.0, 3.0, 6.0	Assumption (only used when needed).	low
level2_time_mult	2.0	1.5, 2.0, 3.0	Assumption (only used when needed).	low
model_A	1.0	0.5, 1.0, 2.0	Fit uncertainty: 9 points from Stim; exponent fixed to 1.	medium
model_pth	1.0	0.8, 1.0, 1.25	Fit uncertainty (Stim threshold ~0.7-1% for this noise model).	medium

Assumptions table. 'Confidence' is our judgement of how well the constant is pinned down; constants marked low are order-of-magnitude. The '+9.2' additive T-count of an earlier draft could not be verified and is not used.

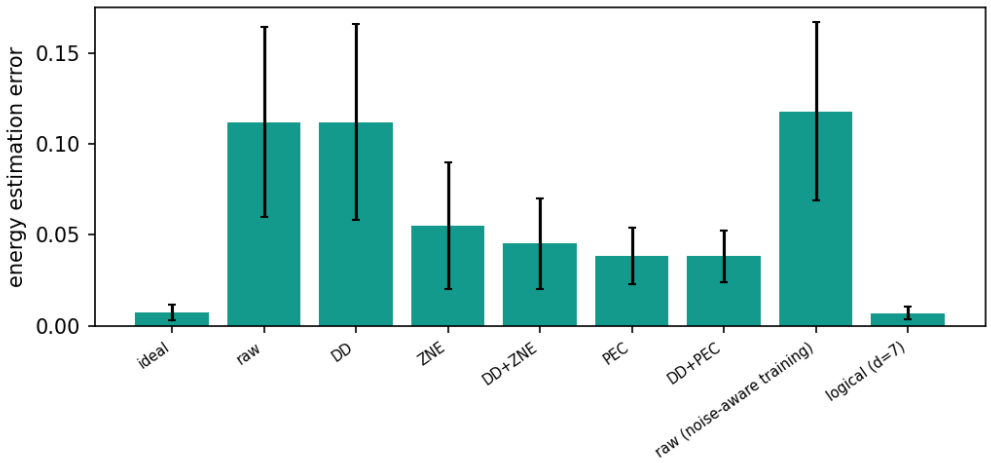
Not modelled: classical decoding latency and reaction time, factory idle errors, non-uniform layouts, measurement-based rotations injected without synthesis (analog-rotation architectures), and correlated physical noise. Estimates are order-of-magnitude tools for scaling arguments, not hardware quotes.

7 (b) Error-mitigation ablation

Instances are QUBOs from random historical dates; each configuration is run with finite shots (2048) on every instance with repeats. *est_err* = |estimated − ideal energy| (estimation accuracy); *gap_norm* = decoded-solution gap normalised by (mean − optimum) (0 = optimal); *p_opt_model* = exact probability of sampling the optimum under the noise model; *overhead* = sampling-variance multiplier of the mitigation.

config	est_err	est_err_sd	gap_norm	hit_opt	p_opt_model	N99	overhead	q_eff
ideal	0.0075	0.0043	0.00153	0.115	6.51e-05	7.53e+04	1	0
raw	0.112	0.0523	0.00174	0.104	5.17e-05	8.86e+04	1	0.0346
DD	0.112	0.054	0.00151	0.115	5.22e-05	8.8e+04	1	0.0332
ZNE	0.0549	0.0347	0.00144	0.125	5.17e-05	8.86e+04	57	0.0346
DD+ZNE	0.0451	0.0248	0.00148	0.125	5.22e-05	8.8e+04	57	0.0332
PEC	0.0383	0.0156	0.00184	0.115	5.17e-05	8.86e+04	12.1	0.0346
DD+PEC	0.0381	0.0143	0.00148	0.125	5.22e-05	8.8e+04	10.7	0.0332
raw (noise-aware training)	0.118	0.0491	0.0017	0.104	5.16e-05	8.85e+04	1	0.0369
logical (d=7)	0.00691	0.00331	0.0013	0.0938	6.51e-05	7.53e+04	1	1.99e-05

Ablation (averages over instances x repeats; N99 = median shots to see the optimum with 99% confidence under the model).



Energy-estimation error by configuration (mean ± sd over runs).

Findings. Raw noise gives est_err 0.112; the best physical-noise mitigation is *DD+PEC* (0.038, overhead 10.7x); the logical-qubit mode reaches 0.007, indistinguishable from ideal (0.007, finite-shot floor). Mitigation reduces estimation bias but barely changes the probability of finding the optimum, which is set by q·n (theory P1): decoding already uses classical evaluation of every sample. ZNE pays a variance factor of 19 per Richardson extrapolation (P4) and PEC pays γ^2 (P3); both overheads grow exponentially in n·q.

8 (c) Out-of-sample results and statistical significance

Walk-forward: at each rebalance the instance is built from past data only (unit-tested: perturbing the future does not change earlier decisions), solved, traded with 10.0 bp costs and held. Test windows: GFC 2008-01-01 to 2009-12-31 and COVID 2019-07-01 to 2020-12-31. Significance: paired stationary-bootstrap test of the Sharpe-ratio difference (mean block 10, 1000 resamples), Wilcoxon signed-rank on per-period returns, Holm correction over all quantum-vs-classical pairs.

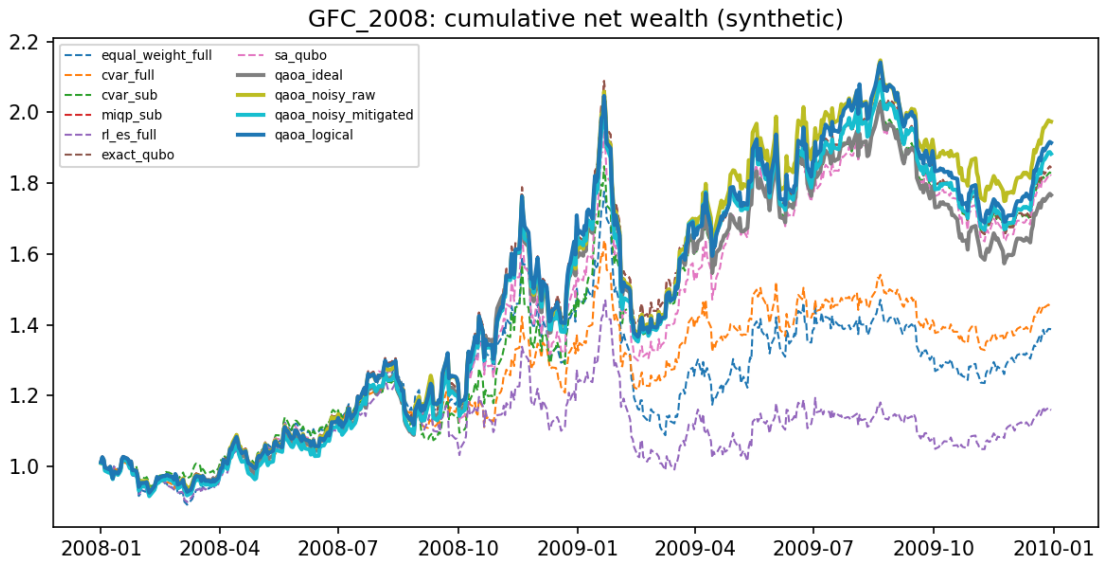
All numbers in this section are from the synthetic market.

Hyper-parameters: GFC_2008: defaults (no tuning run); COVID_2020: defaults (no tuning run).

index	ann_return	ann_vol	sharpe	max_drawdown	avg_turnover	avg_seconds
equal_weight_full	0.171	0.302	0.674	-0.387	0.0703	1.78e-05
cvar_full	0.196	0.249	0.843	-0.257	0.152	0.0105
cvar_sub	0.338	0.282	1.17	-0.267	0.346	0.00774
miqp_sub	0.343	0.308	1.11	-0.328	0.409	0.00056
rl_es_full	0.0738	0.282	0.393	-0.326	0.123	0.000205
exact_qubo	0.343	0.308	1.11	-0.328	0.409	0.00028
sa_qubo	0.336	0.309	1.09	-0.332	0.486	0.0107
qaoa_ideal	0.316	0.31	1.04	-0.328	0.527	1.96
qaoa_noisy_raw	0.388	0.314	1.2	-0.332	0.503	1.45
qaoa_noisy_mitigated	0.357	0.311	1.14	-0.331	0.51	0.973
qaoa_logical	0.368	0.311	1.16	-0.332	0.472	0.895

GFC_2008: out-of-sample performance net of costs.

GFC_2008: 0 of 28 quantum-vs-classical Sharpe differences are significant after Holm correction (bootstrap); 0 by Wilcoxon. Smallest adjusted p = 0.952 (qaoa_noisy_raw vs rl_es_full, difference +0.81, 95% CI [+0.11, +1.59]).



GFC_2008: cumulative net wealth of all methods.

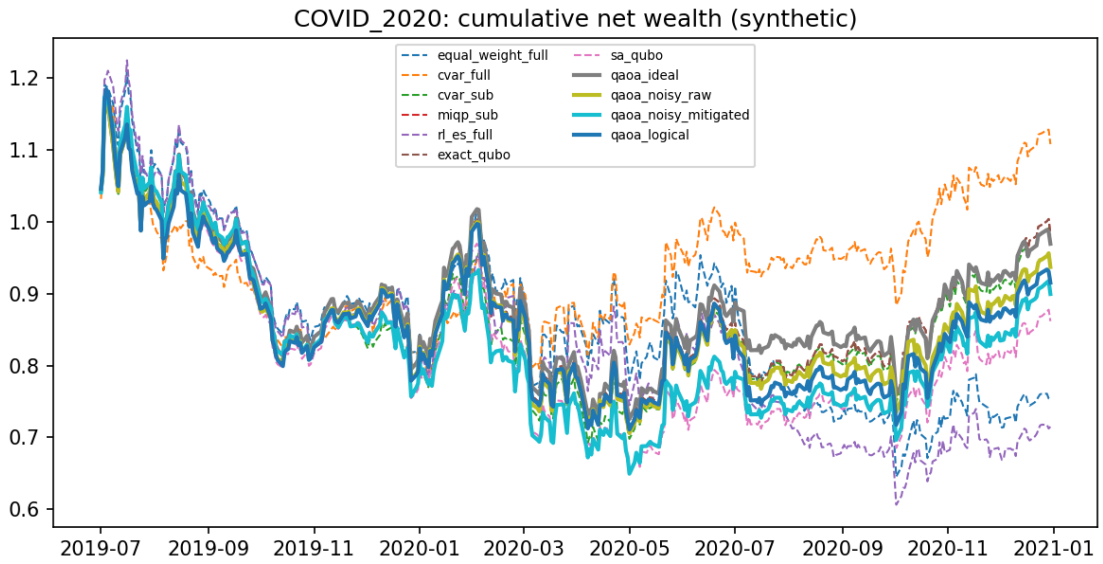
method	gap_norm	approx_ratio	optimum_hit_rate
exact_qubo	-6.05e-17	1	1
qaoa_ideal	0.00157	1	0.2
qaoa_logical	0.00184	1	0.08
qaoa_noisy_mitigated	0.00218	1	0.04
qaoa_noisy_raw	0.00181	1	0.2
sa_qubo	0.0029	1	0.08

GFC_2008: QUBO solution quality per rebalance (hit rate = fraction of rebalances where the exact optimum was returned).

index	ann_return	ann_vol	sharpe	max_drawdown	avg_turnover	avg_seconds
equal_weight_full	-0.167	0.353	-0.341	-0.466	0.0759	1.57e-05
cvar_full	0.068	0.289	0.371	-0.335	0.161	0.00964
cvar_sub	-0.0194	0.322	0.0992	-0.405	0.55	0.00718
miqp_sub	-0.0114	0.325	0.126	-0.391	0.489	0.000582
rl_es_full	-0.194	0.367	-0.404	-0.505	0.377	0.000197
exact_qubo	-0.0114	0.325	0.126	-0.391	0.489	0.000277
sa_qubo	-0.091	0.33	-0.125	-0.448	0.668	0.0104
qaoa_ideal	-0.0198	0.326	0.101	-0.381	0.533	1.9
qaoa_noisy_raw	-0.0408	0.33	0.0378	-0.398	0.515	1.24
qaoa_noisy_mitigated	-0.0657	0.329	-0.0428	-0.447	0.581	0.911
qaoa_logical	-0.0552	0.331	-0.00677	-0.4	0.558	0.471

COVID_2020: out-of-sample performance net of costs.

COVID_2020: 0 of 28 quantum-vs-classical Sharpe differences are significant after Holm correction (bootstrap); 0 by Wilcoxon. Smallest adjusted p = 1.000 (qaoa_ideal vs equal_weight_full, difference +0.44, 95% CI [-0.24, +1.17]).



COVID_2020: cumulative net wealth of all methods.

method	gap_norm	approx_ratio	optimum_hit_rate
exact_qubo	-8.98e-17	1	1
qaoa_ideal	0.00286	1	0.158
qaoa_logical	0.00332	1	0.211
qaoa_noisy_mitigated	0.0047	0.999	0
qaoa_noisy_raw	0.00338	0.999	0.158
sa_qubo	0.00635	0.999	0.0526

COVID_2020: QUBO solution quality per rebalance (hit rate = fraction of rebalances where the exact optimum was returned).

Statistical power. The median half-width of the 95% bootstrap interval for a Sharpe difference is 0.42 (GFC_2008), 0.28 (COVID_2020). Hence Sharpe differences smaller than roughly this size cannot be detected in a single 6-18 month window, and a non-significant result is *absence of evidence*, not evidence of equality. Equivalence (that quantum and classical solvers do not differ) would need the much stronger statement that the interval lies inside a pre-specified margin.

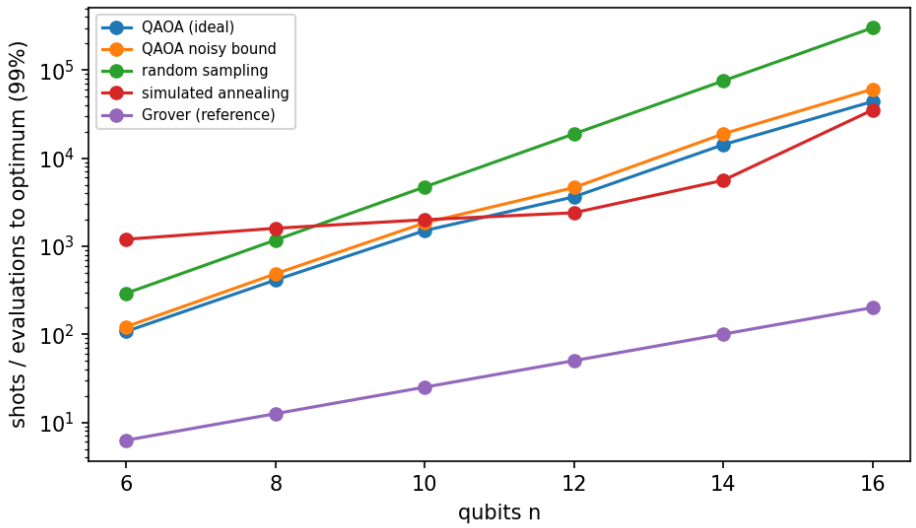
Soft vs hard budget. The exact QUBO optimum coincides with the hard-budget MIQP optimum in 100% of rebalances (GFC_2008); where it does, the QUBO penalty formulation loses nothing and differences between QUBO solvers in P&L are due to solver imperfection, not to the formulation.

Interpretation guardrails. Differences among QUBO solvers reflect how often each returns the exact optimum on a 12-16 qubit problem, not investment skill; Sharpe differences over 6-18 months have wide confidence intervals; the comparison against CVaR and RL conflates the objective (QUBO vs CVaR vs learned policy) with the solver.

9 Scaling study and theory

n	p_layers	p_opt_ideal	N99_qaoa	N99_qaoa_noisy_LB	N99_random	SA_evals_median	grover_queries
6	3	0.0418	108	122	292	1.2e+03	6.28
8	3	0.0111	414	487	1.18e+03	1.6e+03	12.6
10	3	0.00305	1.51e+03	1.84e+03	4.71e+03	2e+03	25.1
12	3	0.00127	3.64e+03	4.64e+03	1.89e+04	2.4e+03	50.3
14	1	0.000324	1.42e+04	1.89e+04	7.54e+04	5.6e+03	101
16	3	0.000105	4.39e+04	6.07e+04	301,802	3.52e+04	201

Shots (QAOA) or single-flip evaluations (SA) to reach the exact optimum with 99% confidence, vs qubits n (medians over instances).



Cost to reach the optimum vs n (log scale). Slopes are fitted exponents in $2^{slope \cdot n}$.

Fitted exponents (cost ~ $2^{s \cdot n}$): N99_qaoa: 0.86, N99_qaoa_noisy_LB: 0.88, N99_random: 1.00, SA_evals_median: 0.43, grover_queries: 0.50. **Caveats:** six sizes, n ≤ 16; p fixed by the adaptive rule; SA 'evaluations' are O(n)-cost single flips while a QAOA shot needs a full cost evaluation and a circuit execution, so the two axes are not directly comparable in wall-clock; SA uses restarts of 200n flips, which quantises its counts. This is trend evidence at toy size, not asymptotics.

Bounds proved for this noise model (docs/THEORY.md, each unit-tested)

- **P1** Success lower bound: $P_{noisy}(z) \geq P_{ideal}(z)(1-q)^n$ for every bitstring z.
- **P2** Energy estimation: $N \geq \Delta^2 \ln(2/\delta)/(2\varepsilon^2)$ shots (Hoeffding).
- **P3** PEC for independent flips: $\gamma = (1-2q)^{-n}$, shots $\propto \gamma^2$; with q = 0.03: $\gamma^2 = 7.2$ at n = 16, 2.4×10^5 at n = 100, 2.8×10^6 at n = 120.
- **P4** ZNE (scales 1,2,3): variance inflation 19 plus truncation bias.
- **P5** Random sampling needs $\ln(0.01)/\ln(1-2^{-n}) \approx 4.6\cdot2^n$ shots; Grover-type $(\pi/4)2^{n/2}$ is a reference line only.
- **P6** Noise budget: a shot-complexity advantage over a classical solver needing N_c evaluations requires $q \leq 1 - (p^*/P_{ideal}(z^*))^{1/n}$, which shrinks like 1/n - the quantitative reason logical qubits (q_L ~ 1e-5) are needed.

Short proofs and a data-derived noise budget

P1. The flip channel is a mixture: with probability $(1-q)^n$ no qubit flips and the output string equals the ideal one, so $P_{noisy}(z) = (1-q)^n P_{ideal}(z) + (\text{non-negative terms}) \geq (1-q)^n P_{ideal}(z)$. **P3.** One flip(q) channel acts on a qubit's distribution as the matrix [[1–q, q],[q, 1–q]] with inverse [[1–q, –q],[–q, 1–q]]/(1–2q), whose absolute entries sum (per column) to 1/(1–2q); for n qubits the quasi-probability norm is $\gamma = (1-2q)^{-n}$. Sampling signs from the inverse gives an unbiased estimator with range $\pm\gamma E_{max}$, hence Hoeffding gives $N = O(\gamma^2\Delta^2 \ln(1/\delta)/\varepsilon^2)$. **P6.** By P1 the shots to see z* with probability 0.99 are at most $\ln(0.01)/\ln(1 - P_{ideal}(z^*)(1-q)^n)$; requiring this to be ≤ N_c and solving for q gives the budget in the text.

n	P_ideal(z*)	classical SA evals	max q for shot advantage	max q x n
6	0.0418	1.2e+03	0.329	1.97
8	0.0111	1.6e+03	0.155	1.24
10	0.00305	2e+03	0.0279	0.279
12	0.00127	2.4e+03	0	0
14	0.000324	5.6e+03	0	0
16	0.000105	3.52e+04	0	0

*Per-qubit flip probability q below which the guaranteed QAOA shot count (P1) stays under the median SA evaluation count, from our own instances. 0 means QAOA already needs more shots than SA with **no** noise. (Shots and SA flips are different units; this is the most generous reading for QAOA.)*

Not proven: any advantage of constant-depth QAOA over the best classical solvers; our scaling data are for n ≤ 16. Tang-style dequantization arguments for mean-variance structure are a further threat to any finance-specific

advantage claim.

10 (d) Interactive dashboard: live re-optimisation under streaming data

`streamlit run dashboard/app.py` replays daily bars tick by tick (synthetic or the real data snapshot); every N ticks it rebuilds the time-dependent QUBO from the data seen so far, re-solves it with each selected method (QAOA variants, exact, SA, MIQP, CVaR, equal weight), and updates P&L lines, target weights, the solver-gap table, the VIX-driven cap and risk aversion. A banner states whether the data are synthetic or real. The streaming source is a replay; swapping in a live poller only requires replacing the bar iterator.

No dashboard screenshots were available when this PDF was built (Streamlit could not run in the build sandbox). Generate them with `python scripts/screenshots.py --real --out results_real` *and rebuild the report.*

11 Software verification

Each claim in this report that can be checked mechanically is covered by a unit test (run with `python -m pytest -q`). The tests are the reason the reported numbers can be trusted to be the numbers the code means to produce:

test	what it establishes
QUBO == direct objective	build_qubo energy equals the hand-written multi-period objective on random bitstrings
QUBO optimum == MIQP optimum	with a large budget penalty the exact QUBO minimum equals the exact hard-budget MIQP optimum (cross-validates both formulations)
MIQP quadratic form	(H, c, const) given to Gurobi reproduces the objective for any integer levels, T = 1..3
QAOA normalised / improves	probabilities sum to 1; trained energy below uniform sampling
Flip channel inverse	apply_inverse_flip_channel exactly undoes apply_flip_channel
PEC exact / unbiased; ZNE reduces bias	infinite-shot PEC equals the ideal energy; sampled PEC is unbiased; ZNE error below raw
Theory P1, P2, P4, P3 values	P1 holds for every bitstring; Hoeffding and Richardson constants; gamma^2 reference values (7.24, 2.37e5)
No look-ahead	changing future returns leaves earlier backtest decisions unchanged
Real-data loaders (mocked)	all yfinance layouts, FRED old/new headers, no back-fill, truncated series dropped, late tickers dropped, pseudo-ESG refused
Ising / dimod conversions	Ising and dict energies equal QUBO energies; Ising phase layer equals the simulator's up to a global phase
Stim suppression; Gurobi == HiGHS; Qiskit == simulator; PyTorch agent	run only where stim / gurobipy / qiskit / torch are installed (skipped otherwise)

Which of the optional tests actually ran for this report is recorded in meta.json (`packages`) and in the status note shipped with the code.

12 Discussion: what a quantum advantage would require

Combining the pieces: (i) P6 says the tolerable per-qubit noise for a shot-complexity advantage decays like 1/n, so any advantage at scale is a logical-qubit statement; (ii) section 6 shows that a logical-qubit execution of 16 qubits needs about 75,576 physical qubits and 0.33 s per shot (about 11 minutes for 2048 shots), and 120 qubits need 58 s per shot (33 hours for 2048 shots), against microseconds for classical exhaustive search at n = 16; (iii) the fitted cost exponent of QAOA (0.86) grows faster than that of simulated annealing (0.43) on our instances, which limits what can be extrapolated; (iv) portfolio problems with realistic constraints (cardinality, round lots, sector limits) have far larger discrete search spaces than our 12-16 qubit encodings, but also much stronger classical solvers (branch-and-bound with convex relaxations, as in the Gurobi MIQP baseline) whose scaling we have not mapped. A credible advantage study would therefore need: instances with n >= 40 where exact methods fail, hardness-calibrated

classical baselines (MIQP at its limits, tuned tabu/SA), QAOA depth scaling, and noise that is correlated and hardware-measured - none of which a 16-qubit statevector study can provide. What this project provides is the machinery and a negative baseline.

13 Threats to validity and conclusion

- **Survivorship bias.** Every ticker is a survivor (Lehman, Bear Stearns, WaMu, Wachovia are absent); this flatters all methods in 2008. The extended universe adds names that fell sharply but survived (C, BAC, AIG, GE, F...), reducing but not removing the bias.
- **Static ESG look-ahead.** ESG scores are a single present-day snapshot applied to 2005-2023; rating agencies also disagree strongly. The hard ESG filter therefore uses information unavailable at the time.
- **Sustainalytics polarity.** Risk scores (lower = better) are converted to a cross-sectional percentile score; the threshold then excludes a fixed fraction of the universe by construction.
- **Adjusted prices and macro data.** Dividends are embedded in adjusted prices; FRED's ICE-BofA credit-spread series is truncated to three years and is dropped, never back-filled.
- **Model-based noise.** NISQ+ noise is a phenomenological flip channel; logical noise is injected from memory simulations; the resource estimator constants are order-of-magnitude.
- **Small n.** At $n \leq 16$ exhaustive search is trivial; all conclusions about scaling are extrapolations from six sizes.
- **Statistics.** Short windows, many pairs (Holm-corrected), and hyper-parameters tuned on windows preceding each test window only; the GFC validation window is calm so it cannot teach crisis behaviour.
- **Untested components.** PyTorch agent not executed; Gurobi MIQP/CVaR not executed.

Conclusion. Rate at which each solver returned the exact QUBO optimum at rebalance time: qaoa_ideal 18%, qaoa_logical 14%, qaoa_noisy_mitigated 2%, qaoa_noisy_raw 18%, sa_qubo 7%. Quantum-vs-classical Sharpe differences significant after Holm correction: GFC_2008: 0 of 28; COVID_2020: 0 of 28. Fitted cost-to-optimum exponents: QAOA 0.86 vs simulated annealing 0.43. The theory shows that noise budgets for an advantage shrink as 1/n and that PEC costs explode with $n \cdot q$, and the resource estimates show that fault-tolerant execution of a 16-qubit instance needs 10^4-10^5 physical qubits. The supported claim is therefore *rigorous bounds plus an empirical test of advantage criteria*; no quantum advantage is demonstrated (and these market numbers are synthetic).

Appendix A: reproduction

```
python scripts/fetch_data.py # internet; writes data/raw
python scripts/build_esg.py # writes data/esg.csv
python scripts/run_all.py --stage qec # needs stim, pymatching
python scripts/run_all.py --stage resources
python scripts/tune.py --real --target GFC_2008; python scripts/tune.py --real --target COVID_2020
python scripts/run_all.py --real --require-optional --stage backtest,ablation,scaling,summary
python scripts/screenshots.py --real --out results_real
python report/build_report.py --results results_real --out report/QPO_Technical_Report.pdf
```

Appendix B: full significance tables

quantum	classical	sharpe_diff	ci_low	ci_high	p_boot	p_boot_holm	p_wilcoxon	p_wilcoxon_holm
qaoa_ideal	equal_weight_full	0.368	-0.2	1.02	0.228	1	0.22	1
qaoa_ideal	cvar_full	0.199	-0.392	0.865	0.552	1	0.426	1
qaoa_ideal	cvar_sub	-0.132	-0.483	0.321	0.514	1	0.0755	1
qaoa_ideal	miqp_sub	-0.0723	-0.241	0.0947	0.392	1	0.178	1
qaoa_ideal	rl_es_full	0.648	-0.107	1.45	0.096	1	0.2	1
qaoa_ideal	exact_qubo	-0.0723	-0.241	0.0947	0.392	1	0.181	1
qaoa_ideal	sa_qubo	-0.049	-0.293	0.222	0.719	1	0.738	1
qaoa_noisy_raw	equal_weight_full	0.529	-0.0244	1.15	0.081	1	0.134	1
qaoa_noisy_raw	cvar_full	0.361	-0.198	0.996	0.235	1	0.134	1
qaoa_noisy_raw	cvar_sub	0.0295	-0.347	0.466	0.892	1	0.833	1
qaoa_noisy_raw	miqp_sub	0.0895	-0.123	0.299	0.383	1	0.465	1
qaoa_noisy_raw	rl_es_full	0.81	0.107	1.59	0.034	0.952	0.0802	1
qaoa_noisy_raw	exact_qubo	0.0895	-0.123	0.299	0.383	1	0.376	1
qaoa_noisy_raw	sa_qubo	0.113	-0.0984	0.333	0.3	1	0.291	1
qaoa_noisy_mitigated	equal_weight_full	0.463	-0.101	1.12	0.156	1	0.22	1
qaoa_noisy_mitigated	cvar_full	0.295	-0.254	0.976	0.365	1	0.113	1
qaoa_noisy_mitigated	cvar_sub	-0.0365	-0.432	0.431	0.876	1	0.958	1
qaoa_noisy_mitigated	miqp_sub	0.0235	-0.164	0.213	0.792	1	0.858	1
qaoa_noisy_mitigated	rl_es_full	0.744	-0.0146	1.59	0.07	1	0.134	1
qaoa_noisy_mitigated	exact_qubo	0.0235	-0.164	0.213	0.792	1	0.858	1
qaoa_noisy_mitigated	sa_qubo	0.0468	-0.205	0.31	0.718	1	0.771	1
qaoa_logical	equal_weight_full	0.488	-0.0737	1.11	0.118	1	0.182	1
qaoa_logical	cvar_full	0.319	-0.227	0.953	0.304	1	0.12	1
qaoa_logical	cvar_sub	-0.0117	-0.42	0.441	0.951	1	0.979	1
qaoa_logical	miqp_sub	0.0484	-0.138	0.231	0.626	1	0.693	1
qaoa_logical	rl_es_full	0.769	0.0257	1.56	0.05	1	0.101	1
qaoa_logical	exact_qubo	0.0484	-0.138	0.231	0.626	1	0.693	1
qaoa_logical	sa_qubo	0.0717	-0.156	0.309	0.574	1	0.819	1

GFC_2008: every quantum-vs-classical pair (bootstrap and Wilcoxon p-values, raw and Holm-adjusted).

quantum	classical	sharpe_diff	ci_low	ci_high	p_boot	p_boot_holm	p_wilcoxon	p_wilcoxon_holm
qaoa_ideal	equal_weight_full	0.442	-0.245	1.17	0.221	1	0.156	1
qaoa_ideal	cvar_full	-0.27	-0.941	0.45	0.44	1	0.395	1
qaoa_ideal	cvar_sub	0.00166	-0.259	0.267	0.991	1	0.984	1
qaoa_ideal	miqp_sub	-0.0251	-0.214	0.16	0.807	1	0.424	1
qaoa_ideal	rl_es_full	0.505	-0.272	1.28	0.202	1	0.145	1
qaoa_ideal	exact_qubo	-0.0251	-0.214	0.16	0.807	1	0.445	1
qaoa_ideal	sa_qubo	0.226	-0.00483	0.484	0.077	1	0.0323	0.906
qaoa_noisy_raw	equal_weight_full	0.379	-0.277	1.1	0.285	1	0.275	1
qaoa_noisy_raw	cvar_full	-0.333	-1.02	0.426	0.329	1	0.332	1
qaoa_noisy_raw	cvar_sub	-0.0615	-0.318	0.206	0.666	1	0.595	1
qaoa_noisy_raw	miqp_sub	-0.0882	-0.278	0.0947	0.375	1	0.0962	1
qaoa_noisy_raw	rl_es_full	0.441	-0.324	1.22	0.273	1	0.241	1
qaoa_noisy_raw	exact_qubo	-0.0882	-0.278	0.0947	0.375	1	0.0962	1
qaoa_noisy_raw	sa_qubo	0.163	-0.075	0.408	0.169	1	0.102	1
qaoa_noisy_mitigated	equal_weight_full	0.299	-0.396	1.04	0.446	1	0.293	1
qaoa_noisy_mitigated	cvar_full	-0.414	-1.13	0.279	0.237	1	0.196	1
qaoa_noisy_mitigated	cvar_sub	-0.142	-0.408	0.119	0.289	1	0.312	1
qaoa_noisy_mitigated	miqp_sub	-0.169	-0.447	0.0627	0.208	1	0.0347	0.936
qaoa_noisy_mitigated	rl_es_full	0.361	-0.415	1.15	0.376	1	0.258	1
qaoa_noisy_mitigated	exact_qubo	-0.169	-0.447	0.0627	0.208	1	0.0347	0.936
qaoa_noisy_mitigated	sa_qubo	0.0821	-0.174	0.308	0.503	1	0.438	1
qaoa_logical	equal_weight_full	0.335	-0.327	1.01	0.356	1	0.275	1
qaoa_logical	cvar_full	-0.378	-1.03	0.332	0.247	1	0.258	1
qaoa_logical	cvar_sub	-0.106	-0.404	0.188	0.473	1	0.891	1
qaoa_logical	miqp_sub	-0.133	-0.355	0.063	0.21	1	0.0546	1
qaoa_logical	rl_es_full	0.397	-0.35	1.18	0.315	1	0.332	1
qaoa_logical	exact_qubo	-0.133	-0.355	0.063	0.21	1	0.0546	1
qaoa_logical	sa_qubo	0.118	-0.193	0.42	0.461	1	0.0684	1

COVID_2020: every quantum-vs-classical pair (bootstrap and Wilcoxon p-values, raw and Holm-adjusted).

Appendix C: resource sensitivity (one parameter at a time)

circuit	parameter	qubits_min	qubits_max	ms_min	ms_max
small n=16	p_phys	16440	282552	152	631
small n=16	synth	56760	75576	109	435
small n=16	routing_overhead	71984	86352	327	327
small n=16	n_factories	26976	202776	70.8	1.31e+03
small n=16	factory_footprint_d2	43176	145776	327	327
small n=16	t_factory_cycles_per_T	75576	75576	178	653
small n=16	cycle_time_us	75576	75576	163	653
small n=16	cnot_parallel_frac	75576	75576	327	327
small n=16	overlap_T_and_clifford	75576	75576	327	328
small n=16	eps_total	56760	97080	244	430
small n=16	level2_footprint_mult	75576	75576	327	327
small n=16	level2_time_mult	75576	75576	327	327
small n=16	model_A	56760	75576	283	327
small n=16	model_pth	56760	97080	283	370
large n=120	p_phys	52308	2057364	1.24e+04	113,220
large n=120	synth	233748	539604	1.01e+04	7.73e+04
large n=120	routing_overhead	486744	698184	5.8e+04	5.8e+04
large n=120	n_factories	304524	1377396	1.31e+04	254,055
large n=120	factory_footprint_d2	349092	952380	5.8e+04	5.8e+04
large n=120	t_factory_cycles_per_T	539604	539604	3.16e+04	115,981
large n=120	cycle_time_us	539604	539604	2.9e+04	115,981
large n=120	cnot_parallel_frac	539604	539604	5.8e+04	5.8e+04
large n=120	overlap_T_and_clifford	539604	539604	5.8e+04	5.8e+04
large n=120	eps_total	233748	647316	2.36e+04	7.09e+04
large n=120	level2_footprint_mult	412596	920628	5.8e+04	5.8e+04
large n=120	level2_time_mult	539604	539604	4.35e+04	8.7e+04
large n=120	model_A	539604	539604	5.8e+04	5.8e+04
large n=120	model_pth	441684	647316	5.25e+04	6.35e+04

Range of physical qubits and time per shot (ms) when each parameter is swept over the values in the assumptions table, others at baseline.

Seeds: Config.seed = 0. Versions: numpy 2.4.4, scipy 1.17.1, pandas 3.0.2. Results folder: results.